

Self-assessment questions Shaping by melting

- In which circumstances shaping by casting is more beneficial than shaping by strain hardening
- Based on learned nucleation and growth theory on solidification which nucleation mechanism and cooling rate condition must be ensured to produce
 - a single crystal
 - a pure metal
- Consider an Al-Si aluminium alloy automotive wheel casting, how both the microstructure of the matrix and eutectic phase can be refined to promote a tough and strong casting
- Is surface free energy strong or negligible in nucleation on cooling
 - Of either pure metals in comparison to alloys ?
 - Of pure metal in either sand mould or metal mould
- Explain the solidification rate phenomena as a function of temperature from 0 K to the melting point
- What is the origin of coring structure
- Which alloying elements in steels are more prone to segregations
- Which factor does promote the columnar structure of casting during solidification of melt contained in a metallic mould
- Which factors do affect segregation during an ingot solidification
- How can regions with large amount of impurities be separated from the ingot during solidification
- Which post-solidification process can eliminate large scale segregations in ingots
- Advantages and disadvantages of MnS in steel castings

Self-assessment questions

- Compare the behaviour of transversal and longitudinal steel texture samples under V-Charpy impact test (CVN)
- How does the CVN of transversal texture compare with the longitudinal one
- Strategies to enhance CVN/ductility in highly textured steel components
- Comment on the simplest test to assess the detrimental presence MnS
- Explain why:
 - a scanning electromagnetic inductor enables the purification of an ingot
 - Spraying a liquid metal on rotating metallic disk promote the formation of amorphous ribbons
 - Impurity segregations are not observed during laser surface remelting
- How do you compare the strength and toughness of casting with that of a strain hardened material
- Following the previous question, how strength and toughness can be improved in either processed materials